Muhammad Muizz Zafar

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Education

University of Ottawa — Ottawa, ON
Bachelor of Applied Science, Software Engineering *GPA: 3.7/4.0* — Sept 2020 - Dec 2024

Work experience

Software Developer — May 2024 - Present
Elections Canada — Ottawa, ON

- Developed **C#/.NET Core LINQ** REST APIs backed by SQL Server databases, enabling internal teams to securely access and consume operational data through reusable endpoints.
- Collaborated with business stakeholders to translate requirements into scalable **ETL solutions** using **Python** and SQL, extracting and transforming data from **Oracle** databases and loading it into a **SQL Server data warehouse**.
- Automated manual processes using **Python**, **powershell**, and **SQL**, for developing HR data workflows to consolidate employee files from PSPC and load data to databases across **DEV, UAT, and PROD** environments.
- Created and maintained **technical documentation** for database releases, ETL workflows, deployment procedures, and production support processes.
- Managed code changes and automated application and database releases using **Python** and **PowerShell** across **DEV, UAT, and PROD** environments through **Git/Azure DevOps**, including branching, pull requests, deployment validation, and production troubleshooting.

Full Stack Developer — September 2023 - April 2024
University of Ottawa — Ottawa, ON

- Developed and maintained full-stack web applications using **Java, Spring Boot, Angular, TypeScript, jQuery, HTML, and CSS**, including an event management and registration platform.
- Redesigned the University of Ottawa's emergency alert website with **Angular** and **Javascript**, introducing a modern, interactive UI. Used by over **20,000** students daily.
- Reduced **Java REST API** response times by shifting filtering and data processing to the database layer and developing optimized **SQL queries**, minimizing unnecessary application-layer processing and data transfer.
- Modernized frontend components using **Angular Material UI** and Javascript to improve usability, and consistency across the application.
- Developed automated unit, integration, and end-to-end tests using **JUnit, Jasmine, and Cypress** to validate backend services, frontend functionality, and regression scenarios.
- Collaborated in an **Agile sprint cycle** to review test results, ensure feature reliability, and **identify and resolve UI/API bugs**.

Cloud Developer Intern — May 2023 - August 2023
Trend Micro Inc. — Ottawa, ON

- Developed serverless monitoring workflows using **AWS Lambda, S3, and CloudWatch** to analyze **50,000+ artifacts per day** and support automated vulnerability scanning pipelines for high-volume cloud artifacts.
- Implemented concurrent logic in **GoLang** using goroutines to improve throughput and strengthen error handling.
- Integrated vulnerability scanning tools (**Anchore, Grype**) with AWS services to automate security checks and operational monitoring.

Cloud Developer Intern — September 2022 - December 2022
Trend Micro Inc. — Ottawa, ON

- Built a serverless automation pipeline using **AWS Lambda, S3, and Athena** to process incoming cloud data events, query metadata, and trigger downstream updates, reducing manual operational effort.
- Configured **Splunk** and **AWS CloudWatch** monitoring, alarms, dashboards, and alerts to track **15+ services** and improve observability of production cloud infrastructure.
- Developed **Docker-based integration tests** and unit tests to improve reliability of distributed serverless data pipelines.

Full Stack Developer Intern — January 2022 - April 2022
Applied Mind Inc. — Ottawa, ON

- Automated event detection by integrating **Python**, **PostgreSQL**, and **GraphQL**, reducing manual detection time.
- Designed real-time dashboards in **Vue.js** for better monitoring and quicker validation of production events.

Full Stack Developer — June 2019 - June 2021
Tech Under Twenty — Oakville, ON

- Created and tested RESTful APIs using **.NET Core (C#)** for the member management system, improving stability and data validation accuracy.
- Designed responsive UI components in **Angular**, performing manual testing and debugging to improve user experience.